ITERATION OF REGULAR INTERACTIONS ON PRESCRIBED COUPLING PREFIXES

Abstract. We prove uniform bounds for two chains of regular interactions formed by prescribed small-field integration steps. The unmarked chain uses fixed physical radii and no inherited remainders. The full chain admits older remainders on running radii. A summable age kernel and a squared-radius remainder sum control their difference uniformly in the length of the prefix. Physical Cauchy estimates then give the marked quadratic action coefficient, with its radius loss retained. For complete profiles with their scalar coefficient subtracted, an exact transport law gives a uniform forward stability estimate. The conclusions assume the actual one-step formation, domain and companion estimates, the stated flow comparisons, and the supplied inherited remainder bounds. They do not construct those inputs or solve for a coupling trajectory.

Contents

1. Introduction 1
2. The regular data at a prescribed step 2
3. Summing the inherited loads 3
4. The two chains and their common step 6
5. The marked estimate 10
6. Extraction on the physical radii 11
7. Forward iteration of complete regular profiles 17
References 23

1. Introduction

A local interaction created by block integration is evaluated again at later scales. Its repeated contribution depends on what has been subtracted from it. A regular family carries its own quadratic action subtraction, and the one-step estimate supplies a decaying weight for its age. A vacuum-subtracted remainder has no such quadratic subtraction. Its bound instead retains the squared ratio of the evaluation radius to its creation radius. The distinction matters when the number of integration steps is unbounded.

We study the iteration on a given finite prefix of reference and running data. The reference data fix the integration coordinates, cutoffs and Gaussian conventions. The action weights vary within their declared complex domains. The two chains use these same choices. Their common part cancels only after the chain on fixed radii has been restricted as the same actual function to the running domain. This identity is stronger than equality of norm bounds or agreement of separately chosen logarithms.

The proof first makes the required one-step hypotheses explicit. It then sums the regular age kernel and the inherited remainder load. A weighted supremum induction bounds the marked difference. The coefficient readout uses the physical background field, so a second derivative costs the inverse square of the physical analyticity radius. Finally, a norm adapted to exact transport gives forward stability for complete regular profiles. This last argument controls the profile for a prescribed coupling sequence; it does not impose the equation determining that sequence.

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Conditional small-field effective-action induction is treated in [3, Theorems 1 and 3, pp. 259 and 264; hypotheses on pp. 260–264]. The local cluster expansion and its effective-action estimates are completed in [4, §2, pp. 11–22]. The arguments below give the finite-prefix summations, domain compatibility and coefficient estimates in the explicitly stated two-chain setting. The marked recurrence is not attributed to those cited theorems.

The inherited remainder estimates, their analytic domains, and the actual one-step theorem are premises of this manuscript. The quantitative statements distinguish the implications proved from these premises from their admission in a full lattice construction. No boundary or large-field history estimate is inferred from a regular-family norm.

2. The regular data at a prescribed step

Fix an odd block factor $L \geq 3$ and a localization side M, a power of L. Use centered compatible grids on the finite four-dimensional tori and their unit-lattice covers. A level-j domain X is a connected union of level-j localization cubes. In level-j units let $d_j(X)$ be the length of a shortest tree in X meeting every cube, divided by M. We use the actual determining windows of the step, including the collars needed for the fluctuation response and for the pointing at unit sites $z \in X$. Every assertion about a common lift includes these windows. Unless a result states a more specific dependence, an unsubscripted C may change from line to line and depends only on the fixed geometry, rates and admitted domain bounds. It is independent of the prefix length, fine mesh and compatible volume. Named input constants remain fixed when the common coupling floor is increased.

The real gauge group is $\mathrm{SU}(2)$ and its complexification is $\mathrm{SL}_2(\mathbb{C})$. Use traceless Hermitian Lie coordinates with pairing $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$, continued bilinearly. The Wilson action is $A(U) = \sum_p (1 - \frac{1}{2}\operatorname{tr} U_p)$ over the stored positive plaquettes, with U_p the oriented plaquette product. The physical current J is a bond one-form in the units of its level.

Write $P = (\mathbf{U}, J)$ for a configuration and current pair on a read window. The current is an independent complex variable in a local core. A core with radii $\mathbf{a} = (a_0, a_1)$ uses representatives $\mathbf{U} = e^{i\eta_j A'} U$ with U real and η_j the bare spacing in level-j units. The radius a_0 bounds the real and complex plaquette defects in units η_j^2 and the current in its level-j units. The radius a_1 bounds A' and its covariant first differences. The real factor must lie in its declared local gauges. The corresponding curvature and current bounds, with the prescribed changes of units and margins, are required at every restricted tower used by the step. Denote the resulting admitted core by $\mathcal{U}_j(X; \mathbf{a})$. We use the core inclusions supplied by the background and one-step theorems. In particular, a substitution along a complex decoupling contour must map into the core of the potential being read, including its original tower and collar data. A radius comparison by itself is not a substitution theorem.

A pointed family G at level j assigns a holomorphic function $G(X, P, z)$ to each (X, z). The value is gauge invariant under the declared action and depends on the pair, independently of its splitting into U, A', J. These analytic families form complex linear spaces. Their real subspaces consist of families that take real values on real admitted pairs. At real values of all input parameters the supplied input and output families belong to these real subspaces. Complex directions and complex parameter values are not subject to that real-valuedness condition. Its norm is

$$\|G\|_{j,\kappa;\mathbf{a}} = \sup_{\mathbb{T},\, X,\, z\in X} e^{\kappa d_j(X)} \sup_{P\in\mathcal{U}_j(X;\mathbf{a})} |G(X, P, z)|. \tag{2.i}$$

Here the supremum includes the admissible tori and the infinite-lattice companion. An unpointed family obeys the same analyticity, gauge and reality conventions on its admitted cores and uses the same norm without z. Parameter suprema are also taken when families are prescribed on a complex parameter domain. There is no sum over z in this norm. The rooted domain sums use the

separate entropy bound of the one-step construction at the chosen decay rate. Restricting a family to a smaller core reduces its norm.

The block map $\mathcal{M}$ is the prescribed one-step average from level-j link fields to level-$(j+1)$ link fields, expressed in their respective units. Its iterates are compositions of these actual maps. For a level-j datum W, let $U_j(W)$ be the supplied fine constrained background with $\mathcal{M}^j U_j(W) = W$, and set $U_0(W) = W$. Its gauge representatives, local analytic continuations and physical current are part of the admitted background data. The block and gauge geometry is developed in [1]; its nonlinear stationary background is an input here.

A companion $\widehat{G}$ is one family on the cover that agrees with each torus function on every common nonwrapping read, after transport of all determining data. We require this agreement for the same frame and weights through the forming step and all compatible cutoffs. Translations preserving the partition act on individual terms. For the real fields with backgrounds $P_j(W) = (U_j(W), J(U_j(W)))$ in the required cores, the pointed total is

$$\mathcal{P}_z(G; W) = \sum_{X \ni z} G(X, P_j(W), z).$$

The scalar Euclidean class requires $\mathcal{P}_{rz}(G; rW) = \mathcal{P}_z(G; W)$ for the admitted lattice isometries, including unit translations, on the common covariance class. This is a property for each base point. The invariance of the sum over base points alone would not give it.

A regular member has this covariance, a companion and its scalar coefficient $b_j(\widehat{G})$. This is the linear readout from the companion's physical Hessian, defined below on the stated response and summability domains. Its own Wilson counterterm is pointed using the same midpoint tents as the regular family. We require the actual fine Wilson action $A(U_j(W))$ to have coefficient one in that readout. Unit-translation covariance of the curl response and affine-curl reproduction by the supplied background give this normalization by the calculation in section 6; their admission is part of the background input. In level-j units, put

$$h_z(x) = \prod_{\mu=1}^{4} (1 - |x_\mu - z_\mu|)_+. \tag{2.ii}$$

Periodize these functions on a torus. The two neighboring linear weights in each coordinate sum to one, so $\sum_z h_z(x) = 1$. Thus the sum of the counterterms is $-b_j(\widehat{G})A(U)$, with A the Wilson action. Each member is first vacuum subtracted. For an unpointed remainder H we subtract only its vacuum value. Its domains are unions of the older cells of side $M\check{R}_j$ at its birth scale. It has a companion and covariant terms under the translations preserving that partition. Its total is invariant on its declared background class under the allowed partition isometries.

3. Summing the inherited loads

Use the fixed blocking factor L and localization side M. Fix an exponent $0 < \alpha_{\mathrm{H}} < 1$. The age of a term of origin j in the step from m to $m+1$ is $n = m + 1 - j$. The regular comparison uses the weight

$$\omega_n = (n+1)^{3/2} L^{-\alpha_{\mathrm{H}}(n-1)}, \qquad n \geq 1. \tag{3.i}$$

This is the weight in the one-step estimate after each regular member's own quadratic action has been subtracted. The use of that estimate includes its stated read domains and the fixed bound for subtracting each member's own counterterm. Its numerical load is charged to the original family norms, as specified below.

Lemma 3.1 (Summability of regular ages)**.** The sum $\Omega = \sum_{n \geq 1} \omega_n$ is finite. It stays finite after multiplying the summand by any fixed nonnegative power of $n+1$. More generally, a supplied kernel bounded by $C(n+1)^d L^{-\alpha(n-1)}$ is bounded by a fixed multiple of eq. 3.i for any $0 < \alpha_{\mathrm{H}} < \min\{1, \alpha\}$. The multiple can depend on $d, L, \alpha - \alpha_{\mathrm{H}}$. ∎

Proof. For each fixed $s \geq 0$ the ratio of successive terms of $(n+1)^s\omega_n$ tends to $L^{-\alpha_{\mathrm{H}}} < 1$. After a fixed age these ratios are bounded by a number less than one. This proves the two summability assertions. For the comparison, the quotient of the proposed majorant and ω_n is $C(n+1)^{d-3/2}L^{-(\alpha-\alpha_{\mathrm{H}})(n-1)}$. The positive exponential reserve bounds this quotient uniformly in n. □

For the running-radius chain fix real numbers $\theta_0, \dots, \theta_N$ above a floor $X > e$ and constants $\lambda_\flat > 0$, $c_5 \geq 0$ and $\bar{B} > 0$. Assume the weak flow comparison

$$\lambda_\flat(j-i) - c_5 \leq \theta_i - \theta_j \leq \bar{B}(j-i), \qquad 0 \leq i \leq j \leq N. \tag{3.ii}$$

For example, when $0 < \lambda_\flat \leq \bar{B}$, the frame $\theta_j = X + \lambda_\flat(N-j)$ meets these inequalities for every finite N. This permits bounded excursions between adjacent entries. The frame is fixed when the action weights vary on their admitted domains. It supplies steps $m+1$ for $0 \leq m \leq N$. For fixed positive $A_1, C_0, C_1, \beta_{\mathrm{s}}$ and $q > \mathfrak{p}_0 \geq 0$, use

$$\alpha_\iota(y) = C_\iota y^{-1/2}(\log y)^q, \qquad \delta(y) = A_1 y^{-1/2}(\log y)^{\mathfrak{p}_0},$$

$$\bar{a}_{m,\iota} = (1+\beta_{\mathrm{s}})\alpha_\iota(\theta_m), \qquad \check{a}_{j,\iota} = \min\{\alpha_\iota(\theta_{j-1}), \alpha_\iota(\theta_j)\}, \tag{3.iii}$$

$$\mathsf{s}_m = \delta(\theta_m)/\bar{a}_{m,0}, \qquad w_{m,j} = \max_{\iota=0,1}(\bar{a}_{m,\iota}/\check{a}_{j,\iota})^2. \tag{3.iv}$$

The enlarged output radius $\bar{a}_{m,\iota}$ is fixed before forming step $m+1$. For an inherited term of origin j, the radius $\check{a}_{j,\iota}$ is a permitted restriction of its creation domain. The weight $w_{m,j}$ is the squared ratio used in the inherited-remainder comparison. Domain admission and the quadratic evaluation estimate that supplies this ratio belong to the one-step input. A comparison of numerical radii alone gives no evaluation identity.

For fixed $r > 0$ set

$$R(y) = \min\{L^b : b \in \mathbb{Z}_{\geq 0},\ L^b \geq (\log y)^r\}, \qquad \check{R}_j = R(\min_{0 \leq i \leq j} \theta_i).$$

An inherited origin $j \leq m$ passes the numerical age gate of step $m+1$ when $M\check{R}_j \leq L^{m+1-j}$. Insertion also requires the supplied companion and analytic admission for that term. Write J_m for the designated existing origins passing the numerical gate, and J_m^{rec} for the existing origins that fail it. These lists are fixed on each parameter domain, since the frame and the existing origin labels are fixed there. Every member of J_m must satisfy its companion and analytic hypotheses throughout that domain. Failure of such a hypothesis makes the step inadmissible there; it does not silently remove that member from the list. Origins in J_m^{rec} are outside the regular exponent of this step.

Lemma 3.2 (Squared-radius loads)**.** Let $\kappa_0 > 4$ and $p = (\kappa_0 - 2)/2$. Assume the creation norms of the inherited remainders are bounded by $\rho_j \leq C_R\theta_j^{-\kappa_0/2}$. For a fixed floor high enough in terms of the stated constants, one has

$$\sum_{j \in J_m} w_{m,j}\rho_j \leq \sum_{j=1}^{m} w_{m,j}\rho_j \leq C_{\mathfrak{h}}\theta_m^{-p}, \qquad C_{\mathfrak{h}} \leq C\left(1 + \frac{1}{\lambda_\flat(p-1)}\right). \tag{3.v}$$

The constant C may depend on $p, q, \beta_{\mathrm{s}}, C_R, c_5, \bar{B}$. For $1 \leq j \leq m$ the radii also satisfy

$$\check{a}_{j,\iota} \leq \bar{a}_{m,\iota} \leq C(m+2-j)^{1/2}\check{a}_{j,\iota}, \qquad \iota = 0, 1. \tag{3.vi}$$

The numerical gate, applied to the full set of origins, selects an initial segment. For its recent complement,

$$\sum_{j \in J_m^{\mathrm{rec}}} w_{m,j}\rho_j \leq C(1 + \log\log\theta_m)^2\theta_m^{-\kappa_0/2}. \tag{3.vii}$$

The constant in this last bound can also depend on r, L, M. All bounds are independent of m, N and the frame above the floor. ■

Proof. The fixed constants C_ι cancel in each ratio in eq. 3.iv. In particular that weight is

$$(1+\beta_{\mathrm{s}})^2 \max_{i\in\{j-1,j\}} \frac{\theta_i}{\theta_m}\left(\frac{\log\theta_m}{\log\theta_i}\right)^{2q}.$$

By the lower bound in eq. 3.ii, both θ_i in this maximum are at least $\theta_m - c_5$. Above a fixed floor the logarithmic ratio is bounded by two. The upper bound for one step gives $\theta_{j-1} \le \theta_j + \bar{B}$. Increasing the floor once more yields

$$w_{m,j} \le C\theta_j/\theta_m \le C(1+m+1-j). \tag{3.viii}$$

In the second inequality use the upper bound from j to m. This also gives the upper inequality in eq. 3.vi. For the lower one take $X - c_5 > e^{2q}$, so each α_ι decreases on $[X-c_5,\infty)$. From $\theta_j \ge \theta_m - c_5$,

$$\check{a}_{j,\iota} \le \alpha_\iota(\theta_m - c_5) \le (1-c_5/\theta_m)^{-1/2}\alpha_\iota(\theta_m).$$

The last factor in front of α_ι is at most $1+\beta_{\mathrm{s}}$ after increasing the fixed floor. The summands in eq. 3.v are therefore bounded by $C\theta_m^{-1}\theta_j^{-p}$. The lower flow comparison gives $\theta_j \ge (\theta_m + \lambda_\flat(m-j))/2$ if $X \ge 2c_5$. The summand $t \mapsto (\theta_m + \lambda_\flat t)^{-p}$ decreases. Its sum over nonnegative integers is at most its value at zero plus its integral on $[0,\infty)$, namely

$$\theta_m^{-p} + \frac{\theta_m^{1-p}}{\lambda_\flat(p-1)}.$$

Multiplication by $C\theta_m^{-1}$ proves the bound. The empty sum at $m=0$ is zero.

For the gate, put $y_j = \min_{i\le j}\theta_i$. If the minimum drops, $y_j - y_{j+1} \le \theta_j - \theta_{j+1} \le \bar{B}$. Above a fixed floor this makes the ratio of $(\log y_j)^r$ to $(\log y_{j+1})^r$ less than L. Rounding up to the next L-power gives $\check{R}_j/\check{R}_{j+1} \in \{1,L\}$. Thus $L^j\check{R}_j$ is nondecreasing. The inequality defining the gate is $ML^j\check{R}_j \le L^{m+1}$, proving the initial-segment property.

If an origin fails the gate, its age $n = m+1-j$ satisfies

$$L^n < M\check{R}_j \le ML(\log\theta_j)^r \le C(\log\theta_m)^r(1+\log(n+1))^r.$$

The last inequality follows from $\theta_j \le \theta_m + \bar{B}(n-1)$ and $\log\theta_m \ge 1$. The last factor is bounded by $C_{L,r}L^{n/2}$ for all $n \ge 1$. Taking logarithms gives $n \le C(1+\log\log\theta_m)$. Each such origin has $\rho_j \le C\theta_m^{-\kappa_0/2}$ by the lower flow bound. Sum $C(1+n)$ from eq. 3.viii over these distinct ages to obtain eq. 3.vii. □

One common geometric margin also follows numerically from these profiles. If the supplied curvature threshold is $\varepsilon(y) = A_0 y^{-1/2}(\log y)^{\mathfrak{p}_0}$ with $A_0 > 0$, then

$$\frac{\check{R}_m^2\varepsilon(\theta_m)}{\alpha_0(\theta_m)} \le \frac{L^2A_0}{C_0}(\log\theta_m)^{\mathfrak{p}_0+2r-q}.$$

Indeed the running minimum is at most θ_m, and rounding gives $\check{R}_m \le L(\log\theta_m)^r$. Thus the stronger choice $q > \mathfrak{p}_0 + 2r$ makes this ratio uniformly small above a fixed floor. This supplies that numerical margin when the one-step geometry requires it. Actual read-domain and collar admission remain separate hypotheses.

The positive lower slope is essential for the aged-load bound. For a constant frame the weight is constant and the number of origins passing the numerical gate grows with the prefix. At $\kappa_0 = 4$ even a linearly decreasing frame can give a logarithmic loss in eq. 3.v. For $\theta_{m-n} = X+n$, $m = X^2$ and X a large integer, take $\rho_j = \theta_j^{-2}$ and sum over the lags $n = m-j$ from X to X^2-1. These origins pass the gate since $M\check{R}_j \le C(\log X)^r < L^{n+1}$. The logarithms of their frame entries are comparable to $\log X$, so their contribution is bounded below by $cX^{-1}\log X$.

4. The two chains and their common step

Let $\theta_0, \dots, \theta_m$ be the supplied real frame prefix, above a common floor. Write x_k for the prescribed Wilson weight and $\tau_k = x_k/\theta_k$. The frame fixes the lattice partitions, Gaussian references, real fluctuation coordinates and tests. The variable x_k does not change those choices. For the absolute chain fix a pair $\mathbf{a}_*$ of small positive radii. Its output and inherited cores use $\mathbf{a}_*$ throughout. In the convention that the one-step theorem enlarges its target radii by $1 + \beta_{\mathrm{s}}$, its target radii before enlargement are $\mathbf{a}_*/(1 + \beta_{\mathrm{s}})$.

For the running chain use the creation radii $\bar{\mathbf{a}}_k$ and the inherited radii $\check{\mathbf{a}}_j$ in eq. 3.iii. The latter are formed only once θ_j is supplied. The inherited restriction uses the same function on the smaller core. We assume the admitted core nesting, including $\bar{a}_{k,\iota} \le a_{*,\iota}$, on the common parameter domains. For both systems the radii satisfy the one-step admissibility conditions, including $CMa_0 < a_1$ and $a_1^2 \le C' a_0$ with the fixed chart constants. If a separate current radius is used, it is subject to the same componentwise domain and ratio tests. The present convention ties it to a_0.

At input step k the full list uses the regular members from levels $1, \dots, k$ and the designated aged remainders H_j at the origins J_k of eq. 3.v. The prefix must supply and admit all its required aged inputs. The inequality $M\check{R}_j \le L^{k+1-j}$ is their numerical age test. Since the lengths are powers of L, the old cell side then divides the new unit side in old units. With centered aligned partitions, the new lattice translations and signed coordinate permutations therefore preserve the old cell partition. This supplies the geometric part of the old remainder's symmetry at insertion. Its background membership still has to hold. A family that fails this age test stays outside the regular list. The unmarked list uses only its own regular outputs, from levels $1, \dots, k$. Both lists begin with no entries.

For either list $\mathfrak{f}$ let $S_{\mathfrak{f}}(U)$ be the sum of its vacuum-subtracted regular families, their own Wilson counterterms and its vacuum-subtracted remainders. The current used in this expression on a physical background is its actual current. Each older entry is evaluated in its own units and on its own determining data. The regular one-step theorem is an input for the following actual finite integral. At real admitted data let $K_{k,\theta_k}(W, dV)$ be the fixed all-small- field kernel on $\mathcal{M}(V) = W$. It uses the given fibre density, gauge weight and fluctuation cut. In the resolved fibre chart its real free array is B and its field is $V'(W, B)$ with $\mathcal{M}(V') = W$. The completed fluctuation logarithm includes the pivot links. The analytic continuations and potential bounds used below refer to that completed map. The admitted background nesting identifies $V'(W, 0) = \mathcal{M}^k U_{k+1}(W)$ and $U_k(V'(W, 0)) = U_{k+1}(W)$ up to a site gauge. Thus gauge-invariant carried differences vanish at zero fluctuation. These identities are required of the supplied backgrounds; their existence is not deduced from the integral formula. Put

$$\begin{aligned}\mathcal{E}_{k+1}[x_k, \mathfrak{f}](W) = \log \int K_{k,\theta_k}(W, dV) e^{-x_k A(U_k(V)) + S_{\mathfrak{f}}(U_k(V))} \\ + x_k A(U_{k+1}(W)) - S_{\mathfrak{f}}(U_{k+1}(W)).\end{aligned} \tag{4.i}$$

The integral retains its prescribed numerical normalization. The regular one-step expansion separates a field-independent scalar $\mathfrak{v}_{k+1}$ from this physical logarithm:

$$\mathcal{E}_{k+1}[x_k, \mathfrak{f}](W) = \mathfrak{v}_{k+1} + \sum_{X,z} E_{k+1}(X, P_{k+1}(W), z).$$

The scalar includes the dimensional and reference-normalizer constants specified by the actual kernel. At the same frozen step and action weight this scalar is independent of the incoming list. The same convention is used on every common read, and this scalar is retained separately in the exact identity. The regular norm below bounds the localized remainder in this identity, not an arbitrary unnormalized logarithm. The scalar has zero curvature coefficient. For a real physical datum, real action weight and real inherited data, the logarithm is the real logarithm of the positive integral.

The reference-weight and weight-shifted Gaussian presentations use the same integral. In its supplied fibre quadratic splitting let Δ_A be the action precision, Δ_G the gauge precision and $\Delta = \Delta_A + \Delta_G$. The weight shift in the action then obeys

$$e^{-\langle B,\Delta B\rangle/2} e^{-(\tau_k-1)\langle B,\Delta_A B\rangle/2} = e^{-\tau_k\langle B,\Delta B\rangle/2} e^{(\tau_k-1)\langle B,\Delta_G B\rangle/2}. \tag{4.ii}$$

If B has dimension d and the scaled Gaussian is normalized, retain the multiplier $\tau_k^{-d/2}$ relative to the normalization at $\tau_k = 1$. Here $\tau_k^{-d/2} = \exp[-(d/2)\log\tau_k]$, with the logarithm continued from zero at $\tau_k = 1$ on the relative disc $|\tau_k - 1| < \sigma_* \le 1/2$. The gauge compensation on the right is still a factor on the cut. The pure Gaussian value of its integral in the action-only split of [2] does not replace it in the interacting integral. On the unrestricted Gaussian with the admitted action–gauge splitting and residual gauge dimension d_{g}, its normalized compensation is $\tau_k^{d_{\mathrm{g}}/2}$ by [2, Proposition 4.3]. The two scalar logarithms are therefore $-(d/2)\log\tau_k$ and $(d_{\mathrm{g}}/2)\log\tau_k$. In the stated decomposition the first belongs to $\mathfrak{v}_{k+1}$, while the second is the physical total of the Gaussian mismatch family. This unrestricted identity leaves the compensation factor on the cut unchanged. All remaining vertices and the cut are kept.

Here are the one-step inputs used by the induction. They concern the same actual kernel and localization on all stated domains.

(1) The real backgrounds and their complex extensions, the actual fibre chart, the completed response and the cuts are supplied on the requested field, coupling and remainder parameter domains. In the prefix parameterization, the maps taking the current parameters to every earlier input's parameters are holomorphic and have images in those inputs' declared domains. For shared prefix coordinates these maps are the coordinate projections. The lists and all discrete choices are held fixed throughout each such domain. In the analytic integral presentation the sharp test is a fixed Borel restriction in the real free variables. The Gaussian matrices and the potentials admit the enlarged decoupling contours of the localization. Their local guards, nonvanishing logarithm domains and simultaneous core inclusions are part of the input. At complete physical data the localized sum, together with the prescribed scalar $\mathfrak{v}_{k+1}$, gives eq. 4.i. At an independent-current pair it is the prescribed local analytic expression.

(2) One prescribed convergent localization gives the pointed output E_{k+1} after the specified scalar has been separated from eq. 4.i. Each localized summand has a determining read and a fixed logarithm branch. The supplied localization preserves the pair gauge identities. On physical data these use the admitted gauge change of variables for the actual fibre measure, relative gauge weight and cut, including the prescribed chart identity if an axial presentation is used. Invariance of an arbitrary gauge-fixing weight is not assumed. Identifications of complete reads transport the formulas, their Borel cuts, branches and their assignment to carriers and points. The one-step pointing identity transports the complete pointed sum, allowing reassignment to carriers under a unit isometry that moves the localization partition. For an admitted isometry r, the real free coordinates change by an orthogonal map $O_{r,W}$ with $V'(rW, O_{r,W}B) = rV'(W,B)$. The real map may depend on W. This change transports the cut, completed response, measure factors and point z to rz. There is an open real fluctuation ball on which all inherited backgrounds have their required covariance. The covariance radii e_j obey $e_{j'} \le L^{2(j'-j)} e_j$ for $j \le j'$. This inequality compares thresholds for the same curvature in different units. The earlier backgrounds used by the actual step also have to lie in their own covariance classes throughout the retained real cut. The creation radius is $c_e\varepsilon(\theta_k)$ at step $k \to k+1$, where ε is the supplied real curvature threshold and c_e is fixed by the domain margin. At every formed level, the supplied background, pair core and companion satisfy the domain, response, summability and cofinal compatibility hypotheses of lemma 6.1 and propositions 6.2 and 6.3 uniformly on the output domains.

The actual gauge and pointing identities supply the symmetry hypotheses as explained below. The readout results then define the new scalar coefficient before this output is used in a later list. Their analytic bounds use the supplied background and family estimates, independently of the present iteration.

(3) For a list of regular entries of norm at most E_0, let $\mathcal{N}_k$ be its sum of the restricted original regular-family norms, before vacuum and quadratic subtraction, with weights ω_{k+1-j}, plus the aged remainder norms with weights $\max_\iota(\bar a_{k,\iota}/\check a_{j,\iota})^2$ in the running chain. The weights ω_n are those in eq. 3.i. The comparison evaluates each regular member with its own counterterm. The fixed subtraction-operator bounds are included in K and in the admission threshold, rather than being treated as contractive. Remainders receive their radius weights without that age gain. For $s(1+\mathcal{N}_k)\le\epsilon_{\mathrm{step}}$, the output is

$$E_{k+1}=N_{k+1}+G_{k+1,\tau_k}+F_{k+1}[\mathfrak{f}],\qquad \|F_{k+1}[\mathfrak{f}]\|\le K\{\delta(\theta_k)+s\mathcal{N}_k+e^{-c\theta_k\delta(\theta_k)^2}\}. \tag{4.iii}$$

The norms are the output norms eq. 2.i with the stated radii. Fixed enlargement factors are absorbed in K and in the threshold for s. Thus s is $\delta(\theta_k)/a_{*,0}$ for the absolute chain and $\delta(\theta_k)/\bar a_{k,0}$ for the running chain. The separately given Gaussian families satisfy $\|N_{k+1}\|\le B_N$ and $\|G_{k+1,\tau}\|\le B_G|1-\tau|$ on the absolute cores. They depend only on the frozen Gaussian data and the weight ratio, and restrict to the same terms on the running cores. The normalization of the first family is fixed by

$$\sum_{X,z}N_{k+1}(X,P_{k+1}(W),z)=\log Z_k(W)-\log Z_k(1),$$

where $Z_k(W)$ is the supplied actual constrained reference Gaussian normalizer without a fluctuation cut. Its logarithm has the fixed branch from the positive real normalizer. This identity identifies its curvature coefficient with that of the physical Gaussian normalization. In the physical action-only splitting the total of the Gaussian mismatch is field independent, by the Gaussian gauge normalization. Its scalar coefficient is zero, although its pointed norm need not have a small-coupling factor.

(4) For two admitted lists on the common smaller cores, the one-step comparison bounds the difference of their actual F outputs by Ks times the weighted load of the differences of their inputs, with the same age and radius weights. This is the comparison for the same completed fibre map, potential and cut domains and logarithm expansion. The admitted marked slot has the prescribed interpolation and contour domains, including the zero-marked list. It is additional to the one-list bound in eq. 4.iii.

The constants in these inputs are uniform over the compatible tori, over the declared parameter domains and over the number of older levels. For every fixed prefix the compatible tori include arbitrarily large periods in every direction, so that each fixed finite read and its collar embed in all sufficiently large members of a cofinal sequence. For the running construction, one sufficient size convention is $P_i\ge 2L^{11}M\check R_i$ at every used level i, where P_i is the shortest torus period in level-i units. Every common lifted read, with its collar, must fit in a box shorter than half each torus side. When forming level j before θ_j is supplied, the known stronger condition $P_j\ge 2L^{11}M\check R_{j-1}$ suffices for these numerical period floors. Indeed $P_i=L^{j-i}P_j$ and $\check R_i\le L^{j-1-i}\check R_{j-1}$ for $i<j$, while an eventual extension has $\check R_j\le\check R_{j-1}$. These size conditions do not supply the analytic read identities. The standalone absolute-radius chain has no remainder-radius condition. The blocking factor, localization side, geometric rules and regular norm cap are fixed before raising the common frame floor. The supplied estimates are uniform over the resulting frame-dependent domains. Existence of these one-step data is the separate small-field result. The conditions here specify the estimates used from it, including the weighted comparison on the enlarged contours.

On the admitted real common covariance class, change B to $O_{r,W}B$. The Wilson term is invariant. The earlier pointed totals, their counterterms and the admitted aged remainder totals transform

as specified. The tents satisfy $h_{rz}(rx) = h_z(x)$. By the one-step pointing identity the transformed rooted expression therefore gives the same pointed value at rz as before at z. A covariance proof for the unpointed integral alone would not give this identity. On a connected complex parameter domain a covariance identity of holomorphic terms continues from an open full-dimensional real slice where both are admitted. For logarithms with already identified exponentials on that domain, a single real anchor fixes the difference in $2\pi i\mathbb{Z}$. This is also the branch check for an inherited remainder whose supplied exponential is invariant. It does not form the remainder.

Proposition 4.1 (Restriction of the same step)**.** Assume the one-step inputs on the absolute and running cores just specified. On every complete common read the unmarked construction on the absolute core restricts to the construction made with its restricted unmarked list on the running core. The full and restricted unmarked outputs admit the marked difference

$$E^{\mathrm{m}}_{k+1} = E_{k+1} - E^{\mathrm{u}}_{k+1}|_{\bar{\mathbf{a}}_k}. \tag{4.iv}$$

The two Gaussian families in eq. 4.iii cancel in this difference. At input level j its regular marked member carries the difference of the two coefficients, and its nonregular inputs are exactly the admitted aged remainders. These identities hold for the companions as well as the torus families. ■

Proof. First compare the unmarked step on the two cores. The initial list is empty. At a later step restrict each earlier unmarked function and its vacuum value by the previous identification. Its coefficient is a readout of the same companion on the common near-zero physical ball. Hence its Wilson counterterm is unchanged. On a common read the actual response and the arguments of these functions agree on the whole fluctuation cut and the prescribed decoupling contours. The same term formula and branch therefore give the same integrand in the real free coordinates. Their integrals and the normally convergent localized series agree. This proves the restriction at the next level. It compares analytic functions on their cores directly. Agreement of totals on the graph of real backgrounds would not identify functions of an independent current.

In the full versus unmarked comparison the two lists give different potentials but have the same reference measure and cuts. Put $D = S_{\mathfrak{f}} - S_{\mathfrak{f}^{\mathrm{u}}}$ on their common reads. Move the last two terms of eq. 4.i inside the exponential. The marked vertex is then exactly

$$D(U_k(V'(W, B))) - D(U_{k+1}(W)).$$

In the localized logarithm put this carried difference in its marked slot and set the slot to zero. The subtracted background value is included once. This is the difference to which the one-step comparison applies; a raw insertion of $D(U_k(V'))$ alone would give a different functional. The prescribed expansion at zero is precisely the restricted unmarked one. At this zero value the two unmarked presentations on physical pairs give the same nonvanishing integral, with the same real logarithm at the reference. Any difference of their two continuations of this logarithm is locally constant with values in $2\pi i\mathbb{Z}$. It vanishes on the connected component containing the reference. On any other admitted common component, agreement uses the matching branch specified by the transported local formulas in the one-step input. The full output uses the prescribed branch at the full marked input. For local independent-current terms the comparison uses the same analytic coefficient expressions and branch choices from the input. This gives eq. 4.iv and cancels N and G_{τ_k}. Linearity of the coefficient and vacuum subtraction gives the list difference with the marked members' own counterterms and all and only the aged remainder inputs.

The same argument compares two tori on a common lifted read. All coefficients transported by that lift use identical data and indices. Choose any compatible torus large enough to contain the lifted read to define the companion there. The comparisons prove independence of that choice. Larger tori exhaust the finite reads. This also proves compatibility with restriction of a frame prefix. Wrapping terms retain their torus formulas and do not define extra companion coefficients. □

Proposition 4.2 (Formation along prescribed prefixes)**.** Assume the preceding one-step inputs uniformly on the prescribed parameter domains. Suppose the regular age weights have finite sum Ω and the aged remainder loads of the running prefixes are bounded by R_*. Fix $E_0 > B_N$ before increasing the common frame floor. Suppose $\delta(\theta) \to 0$, $s \to 0$ in both systems, and the failure term in eq. 4.iii is $O(\delta(\theta))$ as the floor increases. Take a fixed relative coupling radius $0 < \sigma_* \le 1/2$ with $B_G\sigma_* \le (E_0 - B_N)/4$ and with the one-step admission on $|x_k/\theta_k - 1| < \sigma_*$. The running parameters lie in the stated smaller domains. Then a sufficiently high floor, uniform in the prefix length, gives both regular chains from their empty initial lists with norm at most E_0 at every creation. The unmarked output satisfies

$$\|E^{\mathrm{u}}_{k+1} - N_{k+1} - G_{k+1,\tau_k}\| \le C\delta(\theta_k)$$

on its absolute core. The outputs are holomorphic on the admitted parameter domains, causal in the supplied frame and weights, and compatible with the stated torus, companion and core restrictions. The unmarked conclusions also hold on their own under the absolute one-step hypotheses, summable regular weights and the stated tail bound. They require neither a running chain nor inherited remainders nor a lower drift bound; their floor depends only on these absolute inputs. ■

Proof. Put $D_* = \Omega E_0 + R_*$ for the two-chain assertion. For the unmarked assertion alone put $D_* = \Omega E_0$. Choose the floor so that for the relevant systems

$$s(1 + D_*) \le \epsilon_{\mathrm{step}}, \qquad K\{\delta(\theta) + sD_* + e^{-c\theta\delta(\theta)^2}\} \le (E_0 - B_N)/2.$$

These are uniform scalar conditions on the half-line above that floor. All earlier regular creation norms are at most E_0 at the induction step. Their inherited restrictions have no larger norm. Their total load is at most ΩE_0, and the full list adds at most R_*. The one-step hypotheses thus apply to the actual lists. Equation eq. 4.iii gives the new norm at most $B_N + (E_0 - B_N)/4 + (E_0 - B_N)/2 < E_0$. The empty input at the first step starts the same induction. With no remainder terms and fixed absolute radii, $s = \delta/a_{*,0}$ and the bound for F is $C\delta$ as claimed. For example the failure condition follows when $\delta(\theta) = c_\delta \theta^{-1/2}(\log\theta)^{\mathfrak{p}_0}$ with $c_\delta > 0$ and $2\mathfrak{p}_0 > 1$. Its negative exponent then dominates any fixed multiple of $\log\theta$. On running radii the required smallness of s follows from their stated comparison with δ, independently of this failure estimate.

At step k the parameters of the earlier outputs enter only through their already formed analytic functions. The supplied parameter maps land inside those earlier domains, so their compositions are defined and holomorphic. The fixed-domain step and its normally convergent expansion give joint holomorphy with the new parameters. Finite induction gives it on each declared prefix domain. There is no sum over all possible prefixes. The new family, its coefficient and its creation covariance radius use only data through θ_k, x_k. The choice of the next frame entry is needed only to specify the inherited restriction. The common read identities in proposition 4.1 persist at each induction step. For the absolute chain no lower drift bound and no remainder are used. Its independent weight parameters may range over the full admitted product of relative discs. The marked parameter domain projects into that product in the shared weights. For example a tube $|z_i| < \varpi_0\theta_i(\log\theta_i)^{-2q_0}$ with $q_0 \ge 0$, $\theta_i > e$ and $0 < \varpi_0 \le \sigma_*/2$ has this inclusion. Its inherited functions retain their separate declared domains. For the full chain the prescribed weaker flow and smaller analytic domains are used only where required for the running radii and inherited loads. □

5. The marked estimate

Write u_i for the norm of the marked difference of the two outputs at level i, on their common running creation domain, including the prescribed parameter supremum. The level-i output was formed using θ_{i-1}. In later lists it is used by restriction and with its own linear quadratic subtraction. The restriction and subtraction norms used by the one-step comparison are included in its fixed constants.

Proposition 5.1 (Uniform marked recurrence)**.** Use eq. 3.i–eq. 3.iv and the inherited loads of lemma 3.2. Suppose the two constructions start with the empty list and satisfy the one-step comparison and common-restriction hypotheses of proposition 4.1. At every step $m+1$, $0 \le m \le N$, they satisfy

$$u_{m+1} \le K_M \mathsf{s}_m \left(\sum_{i=1}^{m} \omega_{m+1-i} u_i + \sum_{j \in J_m} w_{m,j} \rho_j \right). \tag{5.i}$$

Here K_M is uniform on the stated one-step admission class and is fixed before choosing the frame floor. Above a sufficiently large fixed floor X, one has

$$\sup_{1 \le i \le N+1} \theta_{i-1}^{p} u_i \le 2K_M C_{\mathfrak{h}} \mathsf{s}_*, \qquad u_{m+1} \le 2K_M C_{\mathfrak{h}} \mathsf{s}_m \theta_m^{-p}, \qquad \mathsf{s}_* = \sup_{y \ge X} \frac{\delta(y)}{(1+\beta_{\mathrm{s}})\alpha_0(y)}. \tag{5.ii}$$

If the norms and input estimate hold uniformly on an admitted parameter domain, the bounds are uniform on that same domain. ■

Proof. By proposition 4.1, the two Gaussian families cancel. Linearity of the coefficient and of vacuum subtraction makes the difference of the carried regular actions the action of the marked regular members with their own counterterms. The remaining input differences are the admitted aged remainders. The one-step comparison charges their original family norms, with the fixed subtraction costs included in K_M. It gives eq. 5.i on the same actual cores.

Let $u_m^* = \max_{1 \le i \le m} \theta_{i-1}^p u_i$, with $u_0^* = 0$. For every $i \le m$, the lower flow comparison and the floor give $\theta_{i-1} \ge \theta_m / 2$. Hence

$$\theta_m^p \sum_{i=1}^{m} \omega_{m+1-i} u_i \le 2^p \Omega u_m^*.$$

Multiplying eq. 5.i by θ_m^p and using eq. 3.v gives

$$\theta_m^p u_{m+1} \le K_M \mathsf{s}_m (2^p \Omega u_m^* + C_{\mathfrak{h}}).$$

The ratio defining s_* is a fixed multiple of $(\log y)^{\mathfrak{p}_0 - q}$. Thus we may choose X so that $K_M \mathsf{s}_* 2^p \Omega \le 1/2$. Inductively, if $u_m^* \le 2K_M C_{\mathfrak{h}} \mathsf{s}_*$, the right side is at most $2K_M C_{\mathfrak{h}} \mathsf{s}_m$. This proves both estimates at the new level and starts from $u_0^* = 0$. □

Only frame data through m are used in the estimate of the newly formed output. The later restriction involving θ_{m+1} is performed on that output once this entry is available. The recent load eq. 3.vii bounds the available input sizes in that band. Its use in forming a new remainder or a boundary family requires the separate operations for those outputs.

6. Extraction on the physical radii

The coefficient is a second derivative in the physical block datum. In level-j units write $\eta_j \in (0, 1]$ for the fine mesh, with η_j^{-1} integral, and $V = e^{iB}$ for the unit-lattice block datum. The stored positive bonds specify all signs of the cochain coboundary. In a chart based at the identity pair $P_\circ = (\mathbf{1}, 0)$ use

$$P = (e^{i\eta_j A}, J), \qquad \|(A, J)\|_{1,j;D} = \max\{\|A\|_{\infty,D}, \|\nabla_{\eta_j} A\|_{\infty,D}, \|J\|_{\infty,D}\}. \tag{6.i}$$

The differences in this formula are physical difference quotients on the complete stencils of the declared read window D. The current J is independent in the complex pair domain. On a physical background it is the Wilson current in the physical fine pairing. The two radii r_0, r_1 bound the physical curvature and current, and the C^1 perturbation of the potential, respectively.

Lemma 6.1 (A physical ball in the pair domain)**.** Suppose the hereditary pair domain at radii r_0, r_1 has the following tower input. For a triple whose real and complex plaquette defects are at most $\eta_j^2 \epsilon_0$, whose real factor admits the declared local gauges, whose perturbation has physical C^1 norm at most ϵ_1 and whose independent current is at most ϵ_0, the prescribed real and complex

towers on every allowed restricted window exist and satisfy the hereditary curvature and physical-current bounds with radius $K_T(\epsilon_0+\epsilon_1^2)$ in the respective units. Assume this estimate with uniform constants and on fixed positive ceilings for ϵ_0,ϵ_1. Then the ball

$$\|(A,J)\|_{1,j;D}<a, \qquad a=c\min\{r_0,r_1,1\}$$

about $P_\circ$ lies in the pair domain on D. Here $c>0$ can be chosen independently of the fine mesh, level and compatible volume.

Suppose also that the actual continued physical background pair $p_j(B)$ is holomorphic on $\|B\|_\infty < r_B$, satisfies $p_j(0)=P_\circ$, and has $\|p_j(B)\|_{1,j;D}\le C_B\|B\|_\infty$ in the coordinates of eq. 6.i, with uniform $r_B,C_B>0$. Its data ball of radius $\min\{r_B/2,a/(2C_B)\}$ then lies in the same pair domain and is comparable to a with fixed constants. ■

Proof. Use **1** as the real factor. It has zero curvature and the zero potential in each of the required local gauges. In the product of four link exponentials around a plaquette, the linear term is $i\eta_j^2\mathrm{d}_{\eta_j}A$. Here d_{η_j} is the oriented curl per unit length. The remaining series is bounded by $C\eta_j^2\|A\|_\infty^2$ on the fixed small chart, because every monomial of order at least two contains two factors of size at most $\eta_j\|A\|_\infty$. Thus the plaquette defect is bounded by $C\eta_j^2(a+a^2)$. The independent current is bounded by a. Use $\epsilon_0=C'(a+a^2)$ and $\epsilon_1=a$ in the tower estimate. Reducing c places these numbers below the input ceilings and makes $K_T(\epsilon_0+\epsilon_1^2)<r_0$. It also ensures the simple pair conditions at r_0,r_1. This verifies every hereditary condition on its own window. The assertion about p_j follows by the displayed linear bound on the actual background. The first difference in eq. 6.i is used in the plaquette estimate. □

In the physical chart write H_j for the first derivative of the potential of p_j. The first response of the pair is

$$\mathfrak{h}_jB=(H_jB,\mathrm{d}_{\eta_j}^*\mathrm{d}_{\eta_j}H_jB). \tag{6.ii}$$

Indeed the Wilson action to second order is $\frac12\eta_j^4\sum_p|(\mathrm{d}_{\eta_j}A)(p)|^2$ in four dimensions. Its physical gradient uses the bond pairing with weight η_j^4. Thus the second entry in eq. 6.ii is the derivative of the current on link fields. All these normalizations use the half-trace colour pairing.

6.1. **Polarization of a pointed family.** Use localization domains X of fixed cell side M and denote their tree length in cell units by $d_j(X)$. A base point $z\in X$ is a unit-lattice site. Write X^+ for the fixed-width enlarged read window of X. We use the rooted entropy bound $Ce^{\kappa_D m}$ for the number of domains meeting a given cell with $m\le d_j(X)<m+1$. The diameter in level-j units is at most $CM(m+1)$ and the number of base points is at most $C_M(m+1)^4$. These hypotheses hold on the declared tori and on the companion, with the distance and lift used in the family.

For a pointed family $E(X,P,z)$, let $\mathcal{A}$ be a bound for its weighted supremum norm with exponent κ. Thus each term on its core has modulus at most $\mathcal{A}e^{-\kappa d_j(X)}$. The family is holomorphic in the pair. Use an orthonormal colour basis t_1,t_2,t_3, and write δ_b^s for the unit datum on the positive bond b in colour t_s. For the companion, define

$$\Pi_E(b,c)=\frac13\sum_{s=1}^{3}\sum_{X}\sum_{z\in X}D_P^2E(X,P_\circ,z)[\mathfrak{h}_j\delta_b^s,\mathfrak{h}_j\delta_c^s]. \tag{6.iii}$$

The same definition is used on a torus, with its own physical response. The Hessian is in the coordinates eq. 6.i.

Proposition 6.2 (The radius cost in the polarization)**.** Let $\kappa > \kappa_D$ with a fixed margin. Suppose the pair ball of radius $a > 0$ in eq. 6.i on each read window lies in the core, and the actual first response satisfies

$$\|\mathfrak{h}_j \delta_b^s\|_{1,j;X^+} \le C_h e^{-\delta \operatorname{dist}(b,X^+)} \tag{6.iv}$$

for some fixed $\delta > 0$. The kernel eq. 6.iii converges absolutely and obeys

$$|\Pi_E(b,c)| \le C a^{-2} \mathcal{A} e^{-\delta_1 |b_- - c_-|}. \tag{6.v}$$

Here the distance is periodic for a torus. At a single base point z, the absolute double sum of the corresponding kernel with weight $|b_- - z|\,|c_- - z|$ is bounded by $Ca^{-2}\mathcal{A}$. The same bound, with a constant depending on the powers, holds with any fixed nonnegative powers of $1 + |b_- - z|$ and $1 + |c_- - z|$. The constants depend on the fixed geometry and response and entropy bounds and are uniform in depth and compatible volume. On a finite torus, if the complete pair total is invariant under constant colour conjugations near $P_\circ$, then the kernel is the colour average of the Hessian of the composed physical total. ■

Proof. For two nonzero directions v, w in a read window, take the bidisc with radii $a/(4\|v\|_{1,j;X^+})$ and $a/(4\|w\|_{1,j;X^+})$ in the two direction parameters. Its image lies in the pair ball. The two-variable Cauchy formula yields

$$|D_P^2 E(X, P_\circ, z)[v,w]| \le 16 a^{-2} \mathcal{A} e^{-\kappa d_j(X)} \|v\|_{1,j;X^+} \|w\|_{1,j;X^+}.$$

The estimate is immediate if either direction is zero. On the response directions the two factors contribute $e^{-\delta \operatorname{dist}(b,X^+)}$ and $e^{-\delta \operatorname{dist}(c,X^+)}$. The distance between the lower endpoints is at most the sum of these distances and $C_M(d_j(X)+1)$. Reserve a fixed small fraction of all three exponentials to supply $e^{-\delta_1|b_- - c_-|}$. Choose the fraction so that the remaining tree exponent is larger than κ_D. Root each domain at a cell nearest b. At a fixed root, the domain entropy and the base point count are summable against the remaining tree exponent. The remaining root-distance exponential sums over root cells, also on a torus by counting shortest lifts. This proves eq. 6.v.

For the single base point bound, root at the cell containing z. For a fixed X, group b by a cell of X^+ nearest its lower endpoint. The weight $|b_- - z|$ is at most its distance to that cell plus $C_M(d_j(X)+1)$. The cell count in X^+ and the weighted distance sum are bounded by polynomials in $d_j(X)+1$. Use the same bound for c. The tree exponent exceeds the entropy exponent, so the resulting polynomial factors sum. This proof makes no symmetry assertion about the kernel at a single base point.

For the finite-torus identification, the differential of the pair total at $P_\circ$ is invariant under the adjoint action of SU(2). Differentiating this invariance gives zero on every commutator in each potential and current coordinate. Since commutators span the colour algebra, the differential vanishes. The chain rule for the composed total therefore has zero contribution from the second derivative of the physical background. The remaining term is precisely eq. 6.iii. This argument uses only the colour symmetry of the complete pair total. □

If the families vary holomorphically with admitted parameters and have the same uniform estimates on compact parameter sets, the derivatives and the sums in eq. 6.iii are holomorphic. This follows by the same Cauchy contours and normal convergence. For a given frame the response is fixed. The conclusion also applies to a holomorphic response with the same local bounds.

6.2. **The scalar coefficient and its normalization.**

Proposition 6.3 (The companion coefficient)**.** Assume the hypotheses of proposition 6.2 for the finite families and their companion. On each common nonwrapping lift the actual functions and their pair charts agree with the companion, including their read collars. Assume a cofinal sequence of compatible cubic tori whose periods tend to infinity, and local convergence of the responses on every bounded window along it, with the uniform bound eq. 6.iv. Suppose the finite pair totals are invariant under constant colour conjugations as above, and the full physical totals on these cubic tori are invariant under the unit translations, signed coordinate permutations and physical gauge transformations near the trivial datum. Then the companion kernel is translation invariant, written $\Pi_{E,\mu\nu}(u)$ with $u = b_- - c_-$ and bond directions μ, ν. Its zeroth and first spatial moments vanish. For $\mu \neq \nu$ set

$$b_j(E) = \sum_{u \in \mathbb{Z}^4} \Pi_{E,\mu\nu}(u) u_\mu u_\nu. \tag{6.vi}$$

This value is independent of μ, ν, and

$$\sum_u \Pi_{E,\mu\nu}(u) u_\rho u_\sigma = b_j(E)(\delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho} - 2\delta_{\mu\nu}\delta_{\rho\sigma}). \tag{6.vii}$$

In particular $|b_j(E)| \leq Ca^{-2}\mathcal{A}$. The map b_j is linear and real on real data. It is unchanged by admitted restriction of the same functions around $P_\circ$ and by changing coherent localizations of the same physical total. ■

Proof. Fix two bond arguments and first retain domains of bounded tree length whose nearest root to the first argument lies in a bounded set. Eventually each such window and collar embeds in the torus. Its Hessian at $P_\circ$ agrees with the companion Hessian by the common-chart hypothesis. The responses converge there. The tails of the rooted sums are uniform by the proof of eq. 6.v. Sending both bounds to infinity therefore identifies the infinite kernel with the limit of the physical torus Hessian kernels. Their spatial moments through order two pass to this limit by the exponential bound, using the short representatives on the torus. These steps also show that two coherent localizations of the same total give the same kernel. Restriction of an actual function to a smaller neighborhood of the origin preserves its derivatives.

The covariance identities of the finite physical Hessians pass to the limit. The gauge identity implies that the Hessian annihilates a gradient in either slot. In fact, differentiate the physical gauge identity first in a site gauge and then in a bond datum. At zero the gauge tangent is the ordinary signed site difference, and the derivative of that tangent is multiplied by the zero first differential. The statement passes to the infinite kernel on finitely supported site functions. Colour averaging preserves it.

Write $\widehat{\Pi}(p) = \sum_u e^{-ip\cdot u}\Pi_E(u)$ and $t_\mu(p) = e^{ip_\mu} - 1$. The exponential bound gives derivatives of the symbol near zero, and the Ward identity reads $\widehat{\Pi}(p)t(p) = 0$. Its term of degree one gives $\widehat{\Pi}(0)p = 0$ for every p, hence $\widehat{\Pi}(0) = 0$. Put $U_{\mu\mu}(p) = e^{ip_\mu/2}$ and use the midpoint symbol $U(p)^{-1}\widehat{\Pi}(p)U(p)$. In these coordinates every signed coordinate permutation acts by its ordinary signed matrix on the link directions, since it maps bond midpoints to bond midpoints. Inversion therefore makes this symbol even. This eliminates its linear term, and the changes of phase preserve the degree-two part because the constant and linear parts vanish.

Let $Q(p)$ be that quadratic part. Coordinate reflections force $Q_{\mu\mu} = Ap_\mu^2 + B\sum_{\nu\neq\mu} p_\nu^2$ and $Q_{\mu\nu} = Cp_\mu p_\nu$ for $\mu \neq \nu$. Permutations give common coefficients A, B, C. The leading midpoint Ward vector is ip, so $Q(p)p = 0$. Its μ component gives $A = 0$ and $C = -B$. Thus $Q(p) = B(|p|^2 I - pp^{\mathsf{T}})$. The second derivative of the Fourier series is the negative second moment. The mixed off-diagonal moment in eq. 6.vi is B, and the full tensor is eq. 6.vii. The coefficient bound follows by summing $Ca^{-2}\mathcal{A}|u|^2 e^{-\delta_1|u|}$. Linearity and reality follow from the defining derivatives. □

The limiting response may be taken in any coherent representative family for which the hypotheses of proposition 6.3 hold. Suppose two such responses in the common pair chart differ, on each datum direction B, by $(D_{\eta_j}\lambda_B, 0)$, where the restriction of λ_B to each determining window generates an admitted fine gauge there. For this site-to-link difference, $D_{\eta_j}\lambda(x, x+\eta_j e_\mu) = (\lambda(x) - \lambda(x+\eta_j e_\mu))/\eta_j$. At $P_\circ$ the gauge tangent in the current slot is zero, because the current transforms by conjugation and $J = 0$. Differentiate the gauge identity of a local pair term first in the gauge and then in an arbitrary pair direction. Its first differential at $P_\circ$ vanishes by constant-colour invariance. The resulting Hessian therefore annihilates $(D_{\eta_j}\lambda_B, 0)$ in either slot. Thus both responses give the same term in eq. 6.iii, and hence the same coefficient. This comparison uses actual response convergence for both representative families.

For the unit-lattice Wilson action the quadratic term is $\frac{1}{2}\|\mathrm{d}B\|_2^2$. Its Hessian symbol is $|t|^2 I - tt^*$, whose quadratic part is the Maxwell symbol with coefficient one. This fixes $b_j(A_{\mathrm{block}}) = 1$, with no additional factor of two. For a fine Wilson action pulled back by a background, the same conclusion has a separate hypothesis and warrant. Assume the curl response $\mathrm{d}_{\eta_j} H_j$ is covariant under unit translations and H_j reproduces affine curls exactly in physical units. That is, $\mathrm{d}_{\eta_j} H_j B$ is the constant continuum curl of an affine unit-lattice datum B. Interpret these affine data by the absolutely convergent response moments. The curl of the response to a midpoint plane wave in direction μ then has first-order symbol $ip \wedge e_\mu$ at every fine position in a unit cell. To see this, expand the plane wave at the cell origin. Its constant part has zero curl, its affine part has the stated curl by reproduction, and the second-order remainder is bounded by the second absolute moment of the response. That moment is uniform by eq. 6.iv. Squaring the first-order term in the fine energy and summing with weight η_j^4 over the fine positions in a unit cell gives $|p|^2 I - pp^{\mathsf{T}}$, since the fine weights sum to one for each plaquette orientation. Thus the pulled-back fine Wilson action has coefficient one under this affine response hypothesis.

For the prescribed pointed Wilson insertion the tents sum to one on the physical total. Each tent has bounded support in level units. On a pair core of curvature radius r_0, a fine plaquette action is $O(\eta_j^4 r_0^2)$ by the trace expansion at the identity. The $O(\eta_j^{-4})$ fine plaquettes in a tent give a pointed norm at most Cr_0^2, with the fixed localization weight included in C. When r_0 and a are comparable and the insertion has coefficient one by the preceding normalization, subtracting $b_j(E)$ times this insertion costs at most $C\mathcal{A}$ in the same pointed norm. The scalar normalization uses the total and does not identify the tensor at each base point.

For $g > 0$, if the unit-block datum is written $B = gB_{\mathrm{s}}$, the chain rule gives $D^2(E \circ p_j(g\,\cdot))(0) = g^2 D^2(E \circ p_j)(0)$. The scaled coefficient is $g^2 b_j(E)$, while the Cauchy radius in the scaled datum is proportional to a/g. Its bound is thus $|g^2 b_j(E)| \le C\mathcal{A}(g/a)^2$. Dividing by g^2 gives the same physical estimate $Ca^{-2}\mathcal{A}$. The bound on the physical field cannot be replaced by a bound on the normalized fluctuation variable.

6.3. **The marked coefficient on a prescribed prefix.** At step $k \to k+1$, use the common absolute and running restrictions of the two actual outputs. On the absolute domain the unmarked output decomposes as $N_{k+1} + G_{k+1,\tau_k} + F^{\mathrm{u}}_{k+1}$. Here N_{k+1} is the normalized Gaussian family, and G_{k+1,τ_k} is the family of the change of Gaussian weight. Throughout the admitted physical chart its total is a background-free number by the action–gauge splitting in [2, Proposition 4.3]. Consequently its coefficient vanishes by proposition 6.3. This cancellation uses the splitting identity for the actual Gaussian and does not follow from the size of G_{k+1,τ_k}.

Theorem A (The coefficient bounds on the two domains)**.** Use the pair-domain, response and companion hypotheses of lemma 6.1 and proposition 6.3 uniformly on the corresponding formed families. Suppose the unmarked residual has norm at most

$$C_{\mathrm{u}}\theta_k^{-1/2}(\log\theta_k)^{\mathfrak{p}_0}$$

on its absolute pair core, holomorphically in the independent weight deviations $z_i = \theta_i - x_i$ on $|z_i| < \sigma_*\theta_i$, $0 \le i \le k$. The normalized Gaussian family is fixed on the frame. Suppose the actual marked difference E^{m}_{k+1} in eq. 4.iv has norm at most $C_{\mathrm{m}}\theta_k^{-p}$ on the common running core, where $p = (\kappa_0 - 2)/2$ and $\kappa_0 > 4$. Assume joint holomorphy on the admitted narrower weight tube and inherited remainder domains. The weight tube is contained in $|z_i| < \varpi_0\theta_i(\log\theta_i)^{-2q_0}$ for fixed positive ϖ_0, q_0. Let the frame entries exceed e. The two component radii for that creation step are comparable to $\theta_k^{-1/2}(\log\theta_k)^q$ with fixed positive comparison constants. The absolute radii are fixed positive numbers. Then

$$\begin{aligned} b_{k+1} &= b^0_{k+1} + f_{k+1}(z_0,\dots,z_k) + q_{k+1},\\ |f_{k+1}| &\le C\theta_k^{-1/2}(\log\theta_k)^{\mathfrak{p}_0},\\ |q_{k+1}| &\le C\theta_k^{-(\kappa_0-4)/2}(\log\theta_k)^{-2q}. \end{aligned} \tag{6.viii}$$

The last estimate and its holomorphy hold on the admitted marked parameter domain. The coefficients are real on real data. They are independent of the compatible torus and of subsequent restrictions of the same creation functions. If eq. 5.ii holds on this domain, the bound for q_{k+1} also holds with the extra factor s_k defined in eq. 3.iv. ■□

Proof. Set $b^0_{k+1} = b_{k+1}(N_{k+1})$, $f_{k+1} = b_{k+1}(F^{\mathrm{u}}_{k+1})$ and $q_{k+1} = b_{k+1}(E^{\mathrm{m}}_{k+1})$. Linearity and the vanishing coefficient of G_{k+1,τ_k} give the equality. The absolute core has a fixed physical ball by lemma 6.1, so the extraction bound costs a fixed constant for f_{k+1}. The running core has physical ball radius at least $c\theta_k^{-1/2}(\log\theta_k)^q$ in the same lemma. Its extraction cost is at most $C\theta_k(\log\theta_k)^{-2q}$. Multiplication by θ_k^{-p} gives the stated marked bound. Applying the sharper norm estimate in eq. 5.ii before extraction retains s_k. This spends the physical radius once. When a transferred old quadratic term is charged with the factor $(a/a_i)^2$ on the output radius a, the readout costs $a^{-2}(a/a_i)^2 = a_i^{-2}$. A separate charge of the creation radius would count it twice. The parameter claims follow from the normal convergence in proposition 6.2 and the companion identification in proposition 6.3. The coefficient uses the step-k creation radii and therefore requires no next frame entry. □

Corollary 6.4 (Cauchy bounds in the weight and bare variables)**.** Under theorem A, let ϵ_k denote the unmarked bound in eq. 6.viii and put $\|v\|_\theta = \max_{0\le i\le k}|v_i|/\theta_i$. On the half-polydisc $|z_i| \le \sigma_*\theta_i/2$ one has

$$|Df_{k+1}(\mathbf{z})[v]| \le C_{\sigma_*}\epsilon_k\|v\|_\theta.$$

In particular, when $\theta_i \ge \theta_k - c_5$ on the prefix and $\theta_k \ge 2c_5$, the Lipschitz bound in the unweighted supremum norm is at most $C\epsilon_k/\theta_k$ on that half-polydisc.

For the marked term, suppose the actual fixed-frame outputs and inherited lists are holomorphic along a bare-variable disc $|\zeta - \zeta_0| < R$ with the indicated common domains, cuts, branches and companion. Suppose the marked bound in eq. 6.viii holds throughout this disc. Writing ϵ'_k for that bound, one has

$$|\partial_\zeta q_{k+1}(\zeta)| \le 2\epsilon'_k/R \qquad (|\zeta - \zeta_0| \le R/2).$$

■□

Proof. For nonzero v the line $\mathbf{z}+tv$ remains in the unmarked polydisc for $|t| < \sigma_*/(2\|v\|_\theta)$. Cauchy's formula on that line gives the derivative bound. The zero direction is immediate. Integrating along a segment in the convex half-polydisc gives the Lipschitz bound, and the lower frame comparison

bounds $\|v\|_\theta \le 2\|v\|_\infty/\theta_k$. This argument avoids a sum of bounds for the separate coordinate derivatives. On the marked disc use the actual one-variable holomorphic coefficient, including the inherited dependence. The distance from its half-disc to the outer boundary is at least $R/2$. Cauchy's formula gives the second bound. The line in the unmarked polydisc gives no enlargement of the marked bare-variable disc. □

For example, two inherited lists at real centres of the same frame may be compared on their actual common cores. The list $H_\upsilon = (1-\upsilon)H_0 + \upsilon H_1$ has norm at most $4r$ on $|\upsilon| < 3/2$ if each has norm at most r there before interpolation. The estimate is the triangle inequality. Their common companion and linear covariance identities persist under this interpolation. The list can be used in the marked estimate when its slot admits that norm on those same cores. Holomorphy along this list interpolation and along a bare-variable disc are different hypotheses on the actual inputs.

7. Forward iteration of complete regular profiles

The list estimates also have a formulation for the complete regular profile after its scalar curvature coefficient has been subtracted. This formulation requires a transport estimate on complete functions. An estimate on one chosen local representation alone does not imply the required composition identity.

Fix a finite horizon N. For $0 \le j \le N$, let $\mathcal{B}_j$ be a complex Banach space of complete pointed analytic families at scale j, using the complex classes just defined. Each family has zero value and first derivative at the trivial field, zero companion curvature coefficient, and the prescribed gauge, lattice-symmetry and period-compatibility identities. The spaces use a common physical field radius and a common localization rate. Their norms include the companion and finite-period compatibility bounds. A faithful realization starts with a Banach space of weighted summable local analytic representations, with bounded complete evaluation and compatibility maps. Quotient this Banach space by the joint kernel of all complete evaluations. That kernel is closed, so the quotient is Banach and two quotient classes give the same complete functions only when they are equal. The gauge, covariance, zero-jet and zero-coefficient conditions are imposed as closed identities, using their bounded maps on the quotient. The supplied local formation construction must provide these bounded maps and the compatible analytic representations.

Lemma 7.1 (Affine test of a complete pointing)**.** On the infinite unit lattice let f_z be a complete pointed physical family that is Fréchet holomorphic near zero in the supremum norm on bounded bond data. Assume its physical Hessian kernels have absolutely summable moments of every fixed polynomial order in the distances of their two bond arguments from z, uniformly in z. Define its bulk kernel by the translation sum of these Hessians and its colour-averaged coefficient $b(f)$ by eq. 6.vi. The pair hypotheses of propositions 6.2 and 6.3 are one sufficient source of these moment bounds. Suppose each f_z is gauge invariant, and the family is covariant as a scalar pointing under all unit translations and signed coordinate permutations. Let $F_{\rho\mu} = -F_{\mu\rho}$ be a real constant colour two-form and $a^F_\mu(x) = \frac{1}{2}\sum_\rho (x_\rho - z_\rho)F_{\rho\mu}$ its affine potential. Then the affine test, defined by the absolutely convergent Hessian moments, is

$$\frac{1}{2}D^2 f_z(0)[a^F, a^F] = \frac{1}{2}b(f)\sum_{\rho<\mu}|F_{\rho\mu}|^2. \tag{7.i}$$

■

Proof. The first differential of the complete pointing vanishes by constant colour conjugation. Differentiate its gauge invariance in a site gauge and in a bond datum. Its Hessian annihilates a gradient in either slot. Constant colour conjugation makes every colour block scalar, since the adjoint representation is the full rotation group on the three real colours. Let $K_z(b,c)$ be this scalar Hessian. The assumed absolute moments allow the Ward identities to be tested on gradients of

linear or quadratic site polynomials by cutting them off at radius R and sending R to infinity. The gradients of these cutoffs are bounded by $C(1+|x-z|)$ uniformly in R.

Define the pointed second moment

$$T_{\mu\nu,\rho\sigma}(z)=\sum_{\substack{b \text{ in direction } \mu\\ c \text{ in direction } \nu}}(b_- - z)_\rho (c_- - z)_\sigma K_z(b,c).$$

Testing linear site polynomials gives zero row and column sums in each direction. Testing quadratic site polynomials then makes T antisymmetric under $\mu\leftrightarrow\rho$ and $\nu\leftrightarrow\sigma$. The extra constant in a repeated-index discrete derivative vanishes by the linear test. The same zero sums permit using midpoints in place of the lower endpoints. Translation covariance makes T independent of z. Using the midpoints, a reflection acts on the four indices with the ordinary four sign factors. Thus any nonzero component must have each index occurring an even number of times. Antisymmetry leaves only the two pairings $\mu=\nu,\rho=\sigma$ and $\mu=\sigma,\nu=\rho$ with distinct indices. Permutations and antisymmetry give

$$T_{\mu\nu,\rho\sigma}=t(\delta_{\mu\nu}\delta_{\rho\sigma}-\delta_{\mu\sigma}\delta_{\nu\rho})$$

for one scalar t.

Fix the first lower endpoint at zero in the bulk second moment and sum over z. Translation covariance lets $b_- - z$ range over all first lower endpoints in K_0 and $c_- - z$ over all second ones. The terms involving only the first or only the second endpoint vanish by the directional zero sums. The two cross terms give $-T_{\mu\nu,\rho\sigma}-T_{\mu\nu,\sigma\rho}$. For $\rho=\mu,\sigma=\nu,\mu\neq\nu$ this is t, so $t=b(f)$. Contracting T with $F_{\rho\mu}/2, F_{\sigma\nu}/2$ and multiplying by $1/2$ proves eq. 7.i. □

Proposition 7.2 (Transport after summing old anchors)**.** Fix $i<k$, let $\xi=L^{-(k-i)}$, and suppose the literal intermediate background map B_ξ from final to old datum is Fréchet holomorphic on the admitted fixed radii in the supremum norms on bounded bond data, preserves the zero datum, and intertwines physical gauges and lattice isometries on the scaled old lattice. Assume its first response sends a final affine potential of curvature F to an old affine potential of curvature ξ^2F modulo the gradient of a polynomially bounded site field. This identity includes any node frame change. Put $H_\xi=DB_\xi(0)$ and assume its first absolute row moment is finite,

$$\sup_b\sum_c \|H_\xi(b,c)\|(1+|c_- - \xi b_-|)<\infty,$$

where b is an old bond and c a final bond.

Let $f_{i,z}$ satisfy lemma 7.1, and subtract its own normalized classical pointing to obtain $r_{i,z}=f_{i,z}-b_i(f_i)\mathsf{e}_{i,z}$. Assume the classical pointing also satisfies the hypotheses of lemma 7.1, has coefficient one and its total S_i satisfies the actual identity $S_i(B_\xi(w))=S_k(w)$. The input local transport estimate is the following per-anchor estimate on the composed function $r_{i,z}\circ B_\xi$. It has a normally convergent local representation rooted at the cell of ξz of norm at most $C\varepsilon_i r^2\xi^{4+\beta}$ for $\beta>0$ on fixed output radius and carrier rate. Here $r>0$ is the fixed input radius parameter. Each term depends on the final bond data in a fixed-width determining collar of its connected final-cell carrier. The norm sums the weighted suprema of these terms with the same positive tree weight as the profile norm. Assume the same per-anchor estimate for the companion, and for the finite-period difference with the extra weight $e^{\gamma P/M}$, where P is the final shortest period and $\gamma>0$ is fixed. The companion and finite-period terms use the specified coherent localizations. All constants are independent of i,k,P,z.

Define the new complete pointing by

$$R_{i,k,y}(w)=\sum_z h_y(\xi z)\, r_{i,z}(B_\xi(w)), \tag{7.ii}$$

with periodized tents on compatible tori. Its rooted norm is at most $C'\varepsilon_i r^2 L^{-\beta(k-i)}$, including the stated finite-period compatibility bound. It has zero value and first derivative if $r_{i,z}(0)=0$, and

zero companion curvature coefficient. Its complete pointings preserve the stated covariance. The exact classical decomposition is

$$\sum_{z} f_{i,z}(B_{\xi}(w)) = b_i(f_i)S_k(w) + \sum_{y} R_{i,k,y}(w)$$

on the tori. For several old levels the sum of the remainders has rooted norm at most

$$\frac{C'r^2}{L^{\beta}-1}\sup_{i<k}\varepsilon_i.$$

■

Proof. Write $n = \xi^{-1}$. In one dimension the tent at an integral anchor has samples $1-|m|/n$ for $|m| < n$ and zero otherwise, after translating the index by the anchor times n. Their sum is $1+2\sum_{m=1}^{n-1}(1-m/n) = n$. Products and periodization give the exact quadrature $\xi^4\sum_z h_y(\xi z) = 1$. For each positive weight the old root lies a bounded final distance from y. Adding a bounded join to its carrier multiplies the norm by a fixed constant. Sum the per-anchor estimate with the tent weights and use this quadrature. The same argument applied to the finite-period difference gives the compatibility bound, since the shortest final period is common to all its terms.

The old gauge invariance and covariance of the literal background give gauge invariance of each new pointing. Every final unit translation is an integral old translation in old units, and the tents respect translations and signed coordinate permutations. Finiteness for each fixed transport suffices for the affine identity. It bounds the response to cut-off affine data by $C_{\xi}(1+|\xi b_- - y|)$, uniformly in the cutoff radius, at final anchor y. The old Hessian moments therefore justify the dominated cutoff limit in the affine second derivative. To see why its kernel computes the Hessian on these responses, first take finitely supported final data v with support S. The row bound gives

$$|(H_{\xi}v)(b)| \le \frac{C_{\xi}\|v\|_{\infty}}{1+\operatorname{dist}(\xi b_-, S)}.$$

Thus $H_{\xi}v$ tends to zero at infinity. Its finite truncations converge in supremum norm, and continuity of the old bilinear Hessian identifies its value with the coefficient-kernel sum. Apply this first to finite affine cutoffs and then use the stated moment domination. A unit lattice first-difference norm gives the same topology, since a first difference is bounded by twice the supremum norm.

The first derivative vanishes by the chain rule and the old vanishing first derivative. In the affine second derivative of the composition, the second derivative of the background is therefore multiplied by zero. The first response contributes the old affine curvature $\xi^2 F$. The polynomially bounded gradient disappears by the two-leg Ward identity used in lemma 7.1. Indeed, cut off its site field and use the old Hessian moments to pass to the limit; those moments hold at every fixed polynomial order. The remaining term is the old affine test times ξ^4, and that test is zero since $b_i(r_i) = 0$. The local Hessian and response moments justify each cutoff limit. The finite tent sum preserves this zero at each anchor. In the direct final datum chart write the new rooted representation as $\sum_X F_X$. A term's Hessian vanishes unless both bond directions lie in its determining collar. That collar contains at most $C_M(1+d_k(X))^4$ bonds, each within $C_M(1+d_k(X))$ of the root after the bounded join. For a fixed output ball of radius $r_{\mathrm{out}} > 0$, the bidisc Cauchy estimate therefore bounds its double Hessian moment of total degree a by $C_a r_{\mathrm{out}}^{-2}(1+d_k(X))^{8+a}\|F_X\|_{\infty}$. Summing over X gives every fixed moment in lemma 7.1, because any fixed polynomial is bounded by a constant times the positive exponential tree weight. This uses the weighted sum norm of the representation. Equation eq. 7.i for the output now gives zero companion coefficient.

Sum eq. 7.ii over y and use $\sum_y h_y(\xi z) = 1$ and the assumed exact transport of the classical total. This gives the decomposition. Finally $\sum_{n\ge 1} L^{-\beta n} = (L^{\beta}-1)^{-1}$, which gives the claimed bound for the old levels. Holomorphy and compatibility are preserved by the normally convergent sums on the common output domains. □

Let $T_{i,k} : \mathcal{B}_i \to \mathcal{B}_k$ be bounded linear maps, for $0 \le i \le k \le N$, with

$$T_{i,i} = I, \qquad T_{j,k}T_{i,j} = T_{i,k}, \qquad \|T_{i,k}\| \le C_T\lambda^{k-i}, \qquad C_T \ge 1, \quad 0 < \lambda < 1. \tag{7.iii}$$

The constants and the spaces' analytic and compatibility bounds are independent of N and of the admitted volume. For literal transport, proposition 7.2 supplies the decay bound whenever its per-anchor estimate is uniform in the input profile norm and applies on these common spaces. Its fixed radius factor is included in C_T. The identities below supply composition.

The semigroup requirement can be checked directly when transport is literal pullback followed by the tent pointing of eq. 2.ii. If ξ^{-1} is a positive integer, then

$$\sum_{z\in\mathbb{Z}^4} h_y(\xi z)h_z(t) = h_y(\xi t). \tag{7.iv}$$

In one dimension the function on the right is piecewise linear and its knots are integers. Its linear interpolation from integer values is therefore exact. Taking products proves the displayed identity. Periodizing proves the same identity on compatible tori. If the literal background pullbacks compose up to site gauge, gauge invariance of each complete pointing removes that gauge. Applying eq. 7.iv to the two successive pointing sums then gives $T_{j,k}T_{i,j} = T_{i,k}$. Absolute convergence of those sums is required before changing their order. This argument does not prove the decay estimate in eq. 7.iii.

Lemma 7.3 (A norm adapted to transport)**.** Suppose eq. 7.iii holds. Fix $\lambda < \rho < 1$. For $\psi \in \mathcal{B}_j$ set

$$\|\psi\|_{j,*} = \max_{0\le n\le N-j} \rho^{-n}\|T_{j,j+n}\psi\|_{\mathcal{B}_{j+n}}. \tag{7.v}$$

These are Banach norms satisfying

$$\|\psi\|_{\mathcal{B}_j} \le \|\psi\|_{j,*} \le C_T\|\psi\|_{\mathcal{B}_j}, \qquad \|T_{j,j+1}\psi\|_{j+1,*} \le \rho\|\psi\|_{j,*} \quad (j < N).$$

All constants are independent of the horizon. ■

Proof. The term $n = 0$ gives the lower bound. The transport bound gives $\rho^{-n}\|T_{j,j+n}\psi\| \le C_T(\lambda/\rho)^n\|\psi\|$, which proves the upper bound and completeness. For $j < N$, composition and the change of index $r = n + 1$ give

$$\begin{aligned}\|T_{j,j+1}\psi\|_{j+1,*} &= \rho \max_{1\le r\le N-j} \rho^{-r}\|T_{j,j+r}\psi\|_{\mathcal{B}_{j+r}} \\ &\le \rho\|\psi\|_{j,*}.\end{aligned}$$

□

For the next result the coupling sequence is prescribed independently of the profile recursion. Fix $m > 1$, and write $\varepsilon_{\mathrm{f}}(t) = t^{-1}(\log t)^m$ for $t > 1$. Let $\Theta_0, \dots, \Theta_N > 1$ be a real frame with

$$|\Theta_j - \{\Theta_* + b_\infty(N - j)\}| \le B_a, \qquad b_\infty > 0, \quad B_a \ge 0. \tag{7.vi}$$

Here Θ_* is a lower-scale parameter that will be taken large. The constants b_∞, B_a are fixed. For example, if $\Theta_j = \Theta_* + \sum_{\ell=j}^{N-1} \beta_\ell$ and $|\beta_\ell - b_\infty| \le B_b\lambda_b^\ell$ with $0 < \lambda_b < 1$, then eq. 7.vi holds with $B_a = B_b/(1 - \lambda_b)$. This follows by summing the geometric tail. No positivity of every β_ℓ is needed.

In the regular step, let e_j be the actual classical action pointing, normalized by $b(\mathsf{e}_j^\infty) = 1$ on the companion. For a complete vacuum-subtracted pointing H, its projection is $H - b(H^\infty)\mathsf{e}_j$. On the fixed-radius spaces both this projection and the coefficient readout are assumed bounded with a uniform constant, as supplied by the physical coefficient estimate. They preserve the real subspaces: the readout is real on real families and the actual classical pointing is real on real fields. The seed n_j below is the projection of the Gaussian pointing. The map $\mathcal{F}_j^\perp$ is the projection of the complete non-Gaussian formation, with the scalar vacuum normalization retained separately. Exact transport of the old complete pointing and this projection identify the profile recursion below with the profile part of that step. This identification, including the literal domains and pointings, is part of its one-step admission.

This profile presentation must place the scalar Gaussian weight powers entirely in the vacuum normalization. Constancy of their total alone does not identify each complete pointing with a constant. If the chosen pointing retains a nonconstant family with zero total, that family must be included in $\mathcal{F}_j^{\perp}$ and satisfy its stated bounds. Neither a zero total nor a zero companion coefficient implies those bounds.

Fix $0 < \sigma_{\mathrm{p}} < \sigma_{\mathrm{p}}^{+}$. At step $j < N$ suppose a jointly holomorphic map $\mathcal{F}_j^{\perp}(x,\psi) \in \mathcal{B}_{j+1}$ is given on $|x - \Theta_j| < \sigma_{\mathrm{p}}^{+}\Theta_j$, $\|\psi\|_{j,*} < H_* + 1$. Suppose $n_j \in \mathcal{B}_{j+1}$ and, on this domain,

$$\begin{aligned} \|n_j\|_{j+1,*} &\le B_n, \\ \|\mathcal{F}_j^{\perp}(x,\psi)\|_{j+1,*} &\le C_{H_*}\varepsilon_{\mathrm{f}}(\Theta_j). \end{aligned} \tag{7.vii}$$

These bounds are assumed for the same actual formation map on its buffered domains. The buffer gives the derivative estimates used below, by the following Cauchy argument. Frozen cutoffs, Gaussian scaling and scalar vacuum normalization are part of that map. A bound on an unbuffered domain alone would not give uniform derivative estimates up to its boundary.

Lemma 7.4 (Derivatives on the buffered profile domain)**.** Let X, Y be complex Banach spaces, $H_* \ge 0$, $\Theta > 0$ and $0 < \sigma_{\mathrm{p}} < \sigma_{\mathrm{p}}^{+}$. Suppose $F(x,\psi) \in Y$ is holomorphic on

$$|x - \Theta| < \sigma_{\mathrm{p}}^{+}\Theta, \qquad \|\psi\|_X < H_* + 1,$$

with $\|F(x,\psi)\|_Y \le A$ there. On the inner product of balls $|x - \Theta| \le \sigma_{\mathrm{p}}\Theta$, $\|\psi\|_X \le H_*$, one has

$$\|D_\psi F(x,\psi)\|_{X\to Y} \le 2A, \qquad \|\partial_x F(x,\psi)\|_Y \le \frac{2A}{(\sigma_{\mathrm{p}}^{+} - \sigma_{\mathrm{p}})\Theta}.$$

■

Proof. For any unit vector $v \in X$, the circle $|t| = 1/2$ in the variation $\psi + tv$ lies in the outer profile ball. Cauchy's formula bounds the derivative at $t = 0$ in Y by $2A$. The circle of radius $(\sigma_{\mathrm{p}}^{+} - \sigma_{\mathrm{p}})\Theta/2$ in the variation $x + t$ lies in the outer weight disc. The same formula gives the second estimate. These arguments use one complex line at a time and are independent of the dimension of either space. □

Theorem B (Uniform forward profile bounds)**.** Assume eq. 7.iii, eq. 7.vi and eq. 7.vii. Choose $H_* \ge 1 + 2B_n/(1-\rho)$, where $\lambda < \rho < 1$. There is a floor for Θ_*, depending only on the stated constants, such that every sequence with $|x_j - \Theta_j| \le \sigma_{\mathrm{p}}\Theta_j$ gives a well-defined recursion

$$\psi_0 = 0, \qquad \psi_{j+1} = T_{j,j+1}\psi_j + n_j + \mathcal{F}_j^{\perp}(x_j,\psi_j), \qquad j < N, \tag{7.viii}$$

with $\|\psi_j\|_{j,*} \le H_*$ at every $0 \le j \le N$. The solution is holomorphic on a neighborhood of this closed coupling polydisc. For two such sequences, let $D = \max_{0\le i<N} |x_i - \widetilde{x}_i|$, with $D = 0$ if $N = 0$. Then

$$\|\psi_j(x) - \psi_j(\widetilde{x})\|_{j,*} \le C\frac{(\log\Theta_j)^m}{\Theta_j^2}D, \qquad 0 \le j \le N. \tag{7.ix}$$

The constant and the floor are independent of the horizon and volume. ■□

Proof. By eq. 7.vi, $\Theta_j \ge \Theta_* - B_a$. The function $\varepsilon_{\mathrm{f}}(t)$ tends to zero as t tends to infinity. Choose Θ_* so that the last term in

$$\|\psi_{j+1}\|_{j+1,*} \le \rho H_* + B_n + C_{H_*}\varepsilon_{\mathrm{f}}(\Theta_j)$$

is at most $(1-\rho)H_*/2$. The choice of H_* makes the right side at most H_*. Starting with $\psi_0 = 0$ proves existence by finite induction, with every intermediate profile inside the open formation ball. The same estimate holds on a larger open coupling polydisc whose relative radius is strictly between σ_{p} and σ_{p}^{+}. Finite composition there proves holomorphy.

For stability the coupling disc and profile ball are convex. Apply lemma 7.4 with $A = C_{H_*}\varepsilon_{\mathrm{f}}(\Theta_j)$, and integrate the two derivatives along their straight segments. Absorb the fixed Cauchy constants into C_{H_*}. With $d_j = \|\psi_j(x) - \psi_j(\widetilde{x})\|_{j,*}$ and $k_j = (\log\Theta_j)^m/\Theta_j^2$, this yields

$$d_{j+1} \le \{\rho + C_{H_*}\varepsilon_{\mathrm{f}}(\Theta_j)\}d_j + C_{H_*}k_jD.$$

Increase the floor until the coefficient of d_j is at most a fixed $\rho_1 < 1$. For $i < j$, the frame bound gives $\Theta_i \ge \Theta_j - 2B_a$. On $t > e^{m/2}$ the function $(\log t)^m/t^2$ is decreasing. If $\Theta_i \ge \Theta_j$ this gives $k_i \le k_j$. Otherwise choose Θ_* so $\Theta_j - 2B_a \ge \Theta_j/2 > 1$; then $k_i/k_j \le (\Theta_j/\Theta_i)^2 \le 4$. Iterating from $d_0 = 0$ therefore gives

$$d_j \le C_{H_*}D\sum_{i<j}\rho_1^{j-1-i}k_i \le \frac{4C_{H_*}}{1-\rho_1}k_jD.$$

For $N = 0$ the only profile is zero, and the statement holds directly. □

Corollary 7.5 (The scalar term of the forward profile step)**.** Use the admitted actual step and prescribed sequences of theorem B. Let $\mathcal{A}_j$ be an ambient complex Banach space of complete pointings on the same fixed radii, with zero value and first derivative at the trivial field. Assume its companion readout $H \mapsto b(H^\infty)$ and the projection $\mathsf{P}_jH = H - b(H^\infty)\mathsf{e}_j$ from $\mathcal{A}_j$ to $(\mathcal{B}_j, \|\cdot\|_{j,*})$ are bounded uniformly. The complete classical pointing $\mathsf{e}_j \in \mathcal{A}_j$ has coefficient one and total S_j. Let $\mathsf{N}^{\mathrm{p}}_{j+1}$ be the Gaussian pointing fixed by the reference frame of the actual profile step in $\mathcal{A}_{j+1}$, with scalar vacuum factors kept separately, and put $b^{\mathrm{G}}_{j+1} = b((\mathsf{N}^{\mathrm{p}}_{j+1})^\infty)$. Then

$$\mathsf{N}^{\mathrm{p}}_{j+1} = b^{\mathrm{G}}_{j+1}\mathsf{e}_{j+1} + n_j, \qquad n_j = \mathsf{P}_{j+1}\mathsf{N}^{\mathrm{p}}_{j+1}.$$

Suppose the supplied unprojected non-Gaussian formation $\mathcal{F}_j(x,\psi) \in \mathcal{A}_{j+1}$ is holomorphic on the buffered product used in eq. 7.vii and has norm at most $C_{H_*}\varepsilon_{\mathrm{f}}(\Theta_j)$ there. It uses the same complete pointing and vacuum convention as the actual step, with $\mathcal{F}^{\perp}_j = \mathsf{P}_{j+1}\mathcal{F}_j$. More precisely, after the scalar vacuum factor is separated, the actual exponent on a compatible finite torus is

$$-xS_{j+1} + \sum_y\{(T_{j,j+1}\psi)_y + (\mathsf{N}^{\mathrm{p}}_{j+1})_y + (\mathcal{F}_j(x,\psi))_y\}.$$

This identity is required of the supplied unprojected formation; an identity of its projection alone would not determine the scalar. Set

$$r_j(x,\psi) = b(\mathcal{F}_j(x,\psi)^\infty).$$

On the inner product of balls $|x - \Theta_j| \le \sigma_{\mathrm{p}}\Theta_j$, $\|\psi\|_{j,*} \le H_*$, one has

$$|r_j| + \|D_\psi r_j\| \le C\varepsilon_{\mathrm{f}}(\Theta_j), \qquad |\partial_x r_j| \le C\varepsilon_{\mathrm{f}}(\Theta_j)/\Theta_j. \tag{7.x}$$

The same bounds, in the output profile norm, hold for $\mathcal{F}^{\perp}_j$ and its derivatives. Along two of the prescribed sequences in theorem B,

$$|r_j(x_j,\psi_j(x)) - r_j(\widetilde{x}_j,\psi_j(\widetilde{x}))| \le C\frac{\varepsilon_{\mathrm{f}}(\Theta_j)}{\Theta_j}\max_{0\le i<N}|x_i - \widetilde{x}_i|, \qquad j < N. \tag{7.xi}$$

The scalar action weight carried out of that step is exactly $x_j - b^{\mathrm{G}}_{j+1} - r_j(x_j,\psi_j)$. ■□

Proof. The Gaussian identity is the definition of its projection. The bounded readout and projection give the respective outer norm bounds from that of $\mathcal{F}_j$. Apply lemma 7.4 to each map to obtain eq. 7.x and its profile counterpart. Integrate the two derivatives along the inner weight and profile segments. By eq. 7.ix, the profile difference is at most $C\varepsilon_{\mathrm{f}}(\Theta_j)D/\Theta_j$, where D is the maximum in eq. 7.xi. The additional factor $\varepsilon_{\mathrm{f}}(\Theta_j)$ from $D_\psi r_j$ is bounded above on the admitted half-line. This proves eq. 7.xi.

The old transported profile has zero companion coefficient. In the actual exponent the classical term is $-x_jS_{j+1}$. The Gaussian and non-Gaussian decompositions add respectively $b^{\mathrm{G}}_{j+1}S_{j+1}$ and r_jS_{j+1} to it. The remaining complete pointing is exactly eq. 7.viii. This computes the scalar

carried by one step on the prescribed input. The next prescribed weight is independent of this computation. □

The transport spaces in this section have fixed physical radii. Their norm equivalence does not remove the running-radius factor in quadratic extraction. The theorem constructs profiles for already prescribed couplings. Solving a backward equation for those couplings, or admitting inherited remainder and boundary histories, requires additional results.

References

[1] *Block averaging and gauge geometry for four-dimensional lattice Yang–Mills theory*, companion manuscript, 2026.

[2] *Gaussian block integration and free action-density correlations*, companion manuscript, 2026.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Comm. Math. Phys. **109** (1987), 249–301. doi:10.1007/BF01215223.

[4] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Comm. Math. Phys. **116** (1988), 1–22. doi:10.1007/BF01239022.